

LABORATORY 1: REVIEW OF DIODES AND CHARACTERISTICS

1. OBJECTIVE

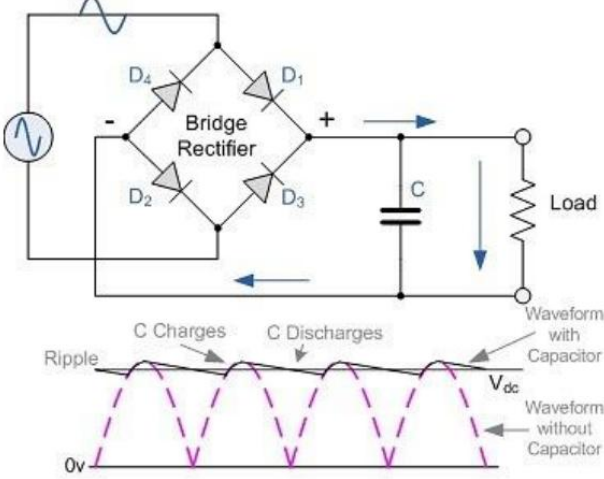
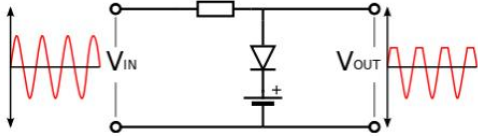
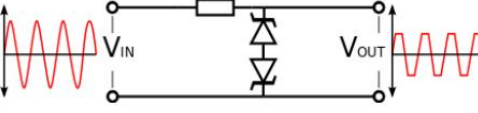
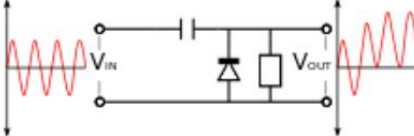
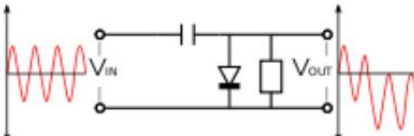
Review the concepts, characteristics and the application of diodes.

2. BACKGROUND

A diode is a two-terminal electronic component that conducts current primarily in one direction (asymmetric conductance); it has low (ideally zero) resistance in one direction, and high (ideally infinite) resistance in the other. This unidirectional behavior is called rectification, and can be used to regulate voltages (e.g., AC to DC, clipping and/or clamping, etc.). Zener diodes (or avalanche diodes) can protect circuits from high voltage surges due to their breakdown feature, while light-emitting diodes (LEDs) can illuminate.

Key knowledges and formulas regarding to diodes applications.

I-V curve of a diode (i.e., PN junction): $I = I_s \left(e^{\frac{V_D}{V_T}} - 1 \right)$ $V_T = \frac{kT}{q}$	
Diode models (ideal and linear) in electrical circuits:	
A Zener diode model:	
A half wave rectifier:	

A full wave bridge rectifier:	
Clipping Circuits:	 A positive peak clipper circuit.  A two-shunt diode clipper circuit.
Clamping Circuits	 A positive unbiased clamp.  A negative unbiased clamp.

3. LAB DELIVERIES

PRELAB:

- (1) Review the knowledge of regular diodes, rectifiers and Zener diodes, which are listed in the previous section.
- (2) Overview the key character of diodes in their datasheets.

1N4001:

<https://www.vishay.com/docs/88503/1n4001.pdf>

1N4148:

<http://www.onsemi.com/pub/Collateral/1N914-D.PDF>

1N4735:

<https://www.mouser.com/datasheet/2/427/1n4728a-104110.pdf>

1N750:

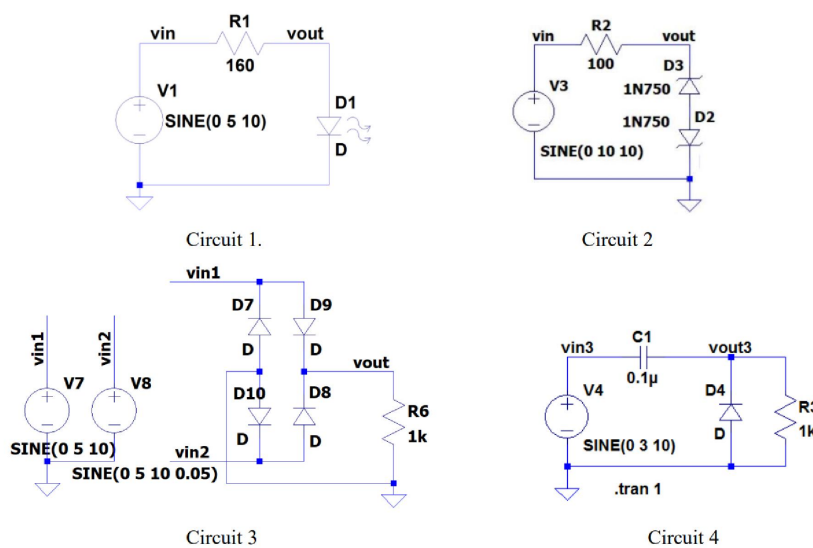
<https://media.digikey.com/pdf/Data%20Sheets/Fairchild%20PDFs/1N746A-759A,4370A-4372A.pdf>

(3) Use LTspice to simulate Circuit 1 & Circuit 2.

1) Observe the voltage at “vin” and “vout” in Circuit 1. Write down the peak voltage of “vout”.

What is the threshold voltage of the LED according to the waveform (voltage drop)?

2) Observe the waveform in Circuit 2. What is the threshold voltage and breakdown voltage of this Zener diode according to the waveform (voltage drop)? Compare with 1N750 in datasheet.



(4) Use LTspice to simulate Circuit 3.

1) Observe the voltages and waveforms of “vin1-vin2” and “vout”. “vin1” should be the reverse of “vin2”, and the same common ground of the circuit. What function does this circuit achieve?

2) Repeat 1) with a 0.22 uF capacitor connected at load in parallel. Explain the changes.

3) Repeat 2) but use Zener diode in reverse orientation, with load $R = 1\text{k}\Omega$, $C = 0.22\mu\text{F}$.

(5) Use LTspice to simulate Circuit 4.

1) Observe the voltage at “vin3” and “vout3”. Write down the peak voltage (V_p) and peak-to-peak voltage (V_{pp}) of “vout3”.

2) Change the capacitor to 100 uF. Observe the changes at output.

3) Stay with 100 uF, change the input frequency, and run the ac analysis for Bode plot from 10Hz to 1MHz. What can you conclude?

4. REFERENCES & ACKNOWLEDGEMENT

(1) Adel S. Sedra & Kenneth C. Smith, “Microelectronic Circuit”, 6th Ed.

(2) B. Razavi, Fundamentals of Microelectronics, 1st ed., John Wiley & Sons Inc., 2008, ISBN:978-0471478461

(3) <https://en.wikipedia.org/wiki/Diode>

(4) Related diode datasheets.